

Week 5 – Backend Optimization & API Improvements

Summary:

Week 5 focused on optimizing the backend for **faster code execution** and enhancing API endpoints for better stability and scalability.

Key Activities:

- Analyzed performance bottlenecks in Python, C, C++, and Java execution pipelines.
- Implemented **async API endpoints** in FastAPI for concurrent execution.
- Added **execution time logging** and resource monitoring for each code run.
- Enhanced error handling for long-running or resource-intensive programs.

Technologies Used: Python, FastAPI (async), WSL2, Docker, GCC, g++, JDK

Outcome:

Backend is optimized for **concurrent multi-language execution**, providing faster response times and stable performance under load.

Flowchart:

